

Shared Sensemaking and Group Cognition: How Teams Think Together

Executive Summary

Teams don't just share information — they build **shared mental models** that enable coordinated understanding and rapid decision-making. Under Active Inference, team cognition is the process of aligning individual generative models into a collective model that all members use for prediction and action. This module examines team mental model alignment, the value and risk of cognitive diversity, groupthink as a precision pathology, and how dialogue and deliberation processes build shared cognition.

Learning Objectives

1. Define **shared mental models** as aligned generative models across team members
 2. Explain how **cognitive diversity** (different models) and **cognitive alignment** (shared models) are both necessary
 3. Identify **groupthink** as excessive alignment — when cognitive diversity is suppressed
 4. Design **dialogue processes** that build shared cognition without destroying diversity
 5. Apply tools for assessing and improving team cognition
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Key Concepts

1. Shared Mental Models

A shared mental model is the portion of the generative model that team members hold in common — the agreed-upon understanding of the situation, the task, and each other's roles:

Case Study — Firefighting Crews: Karl Weick's analysis of the Mann Gulch disaster showed that when a firefighting crew's shared mental model collapsed (they could no longer predict

each other's behavior), the team disintegrated and individuals scattered. Collective cognition depends on shared models — when they break, the team ceases to function.

2. Cognitive Diversity vs. Alignment

State	Team Characteristic	Benefit	Risk
High diversity, low alignment	Members have different models and can't coordinate	Novel ideas, broad coverage	Chaos, miscommunication, paralysis
Low diversity, high alignment	Members share one model and think alike	Speed, efficiency, cohesion	Groupthink, blind spots, fragility
High diversity, high alignment on process	Members have different models but share protocols for integrating them	Innovation + coordination	Requires skilled facilitation

3. Groupthink as Precision Pathology

Groupthink occurs when the team assigns excessive precision to its consensus model and very low precision to any dissenting observations. The mechanism: social pressure makes disagreement costly → individuals suppress private information → the team's shared model becomes an unfalsifiable echo chamber.

Prevention strategies:

- Assign a "red team" or devil's advocate
- Create psychological safety for dissent
- Use anonymous input mechanisms
- Rotate members to prevent excessive homogeneity

4. Dialogue and Deliberation

Effective team cognition requires structured conversation:

Process	Purpose	When to Use
Brainstorming	Divergent idea generation	Early-stage exploration
Structured debate	Testing competing models	When options must be evaluated
Consensus building	Aligning on shared model	When coordination requires agreement
After-action review	Model updating from experience	After any significant event
Pre-mortem	Testing model assumptions	Before committing to action

Cross-References

- For organizational mental models, see [Organizational Systems: Cognition](#)
- For strategic assumption audit, see [Strategic Modeling: Cognition](#)
- For ML inference and hybrid cognition, see [Digital Transformation: Cognition](#)

Summary

Concept	Team/Collective Meaning
Shared mental models	The aligned generative models team members hold in common
Cognitive diversity	Different models across members — broadens coverage
Cognitive alignment	Shared models that enable coordination
Groupthink	Excessive alignment that suppresses diversity
Dialogue processes	Structured conversations for building shared cognition

References

- Weick, K. E. (1993). The collapse of sensemaking in organizations: The Mann Gulch disaster. *Administrative Science Quarterly*, 38(4), 628–652.
- Janis, I. L. (1982). *Groupthink*. Houghton Mifflin.
- Cannon-Bowers, J. A., & Salas, E. (2001). Reflections on shared cognition. *Journal of Organizational Behavior*, 22(2), 195–202.